

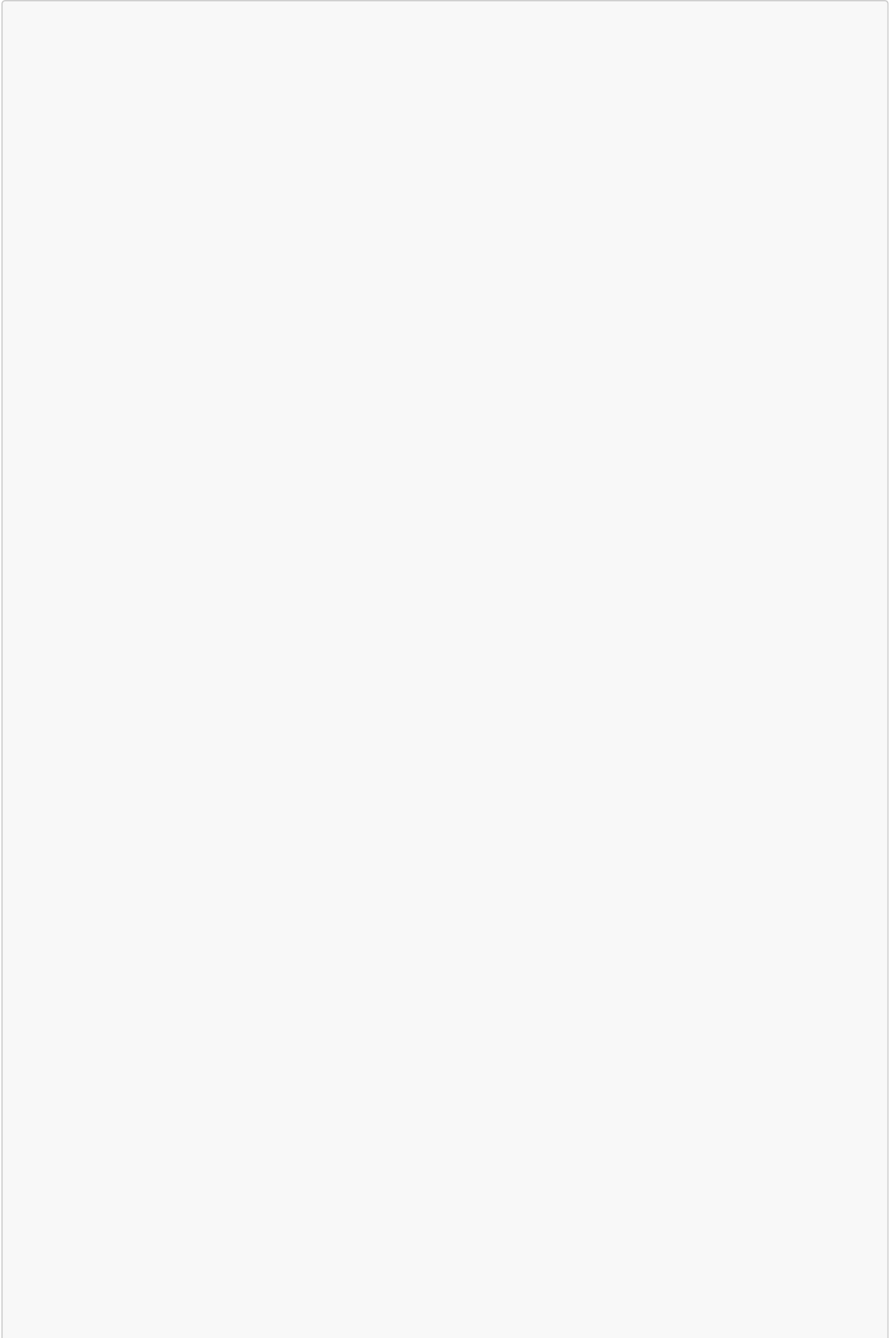
TECHNICAL DOCUMENT 4: TEST AUTOMATION UML DIAGRAMS & DOMAIN SPECIFICATION

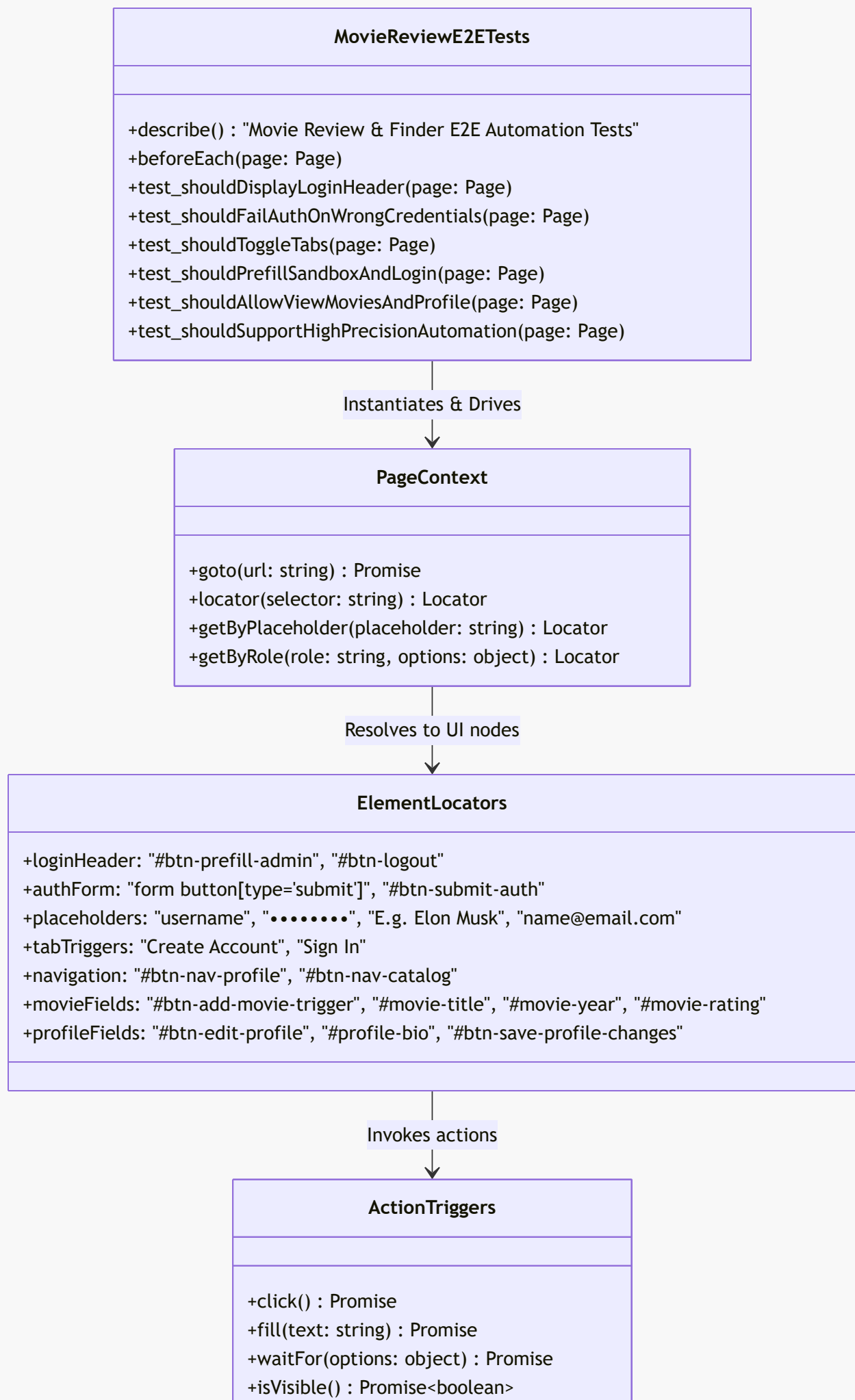
Movie Review & Finder Web Application

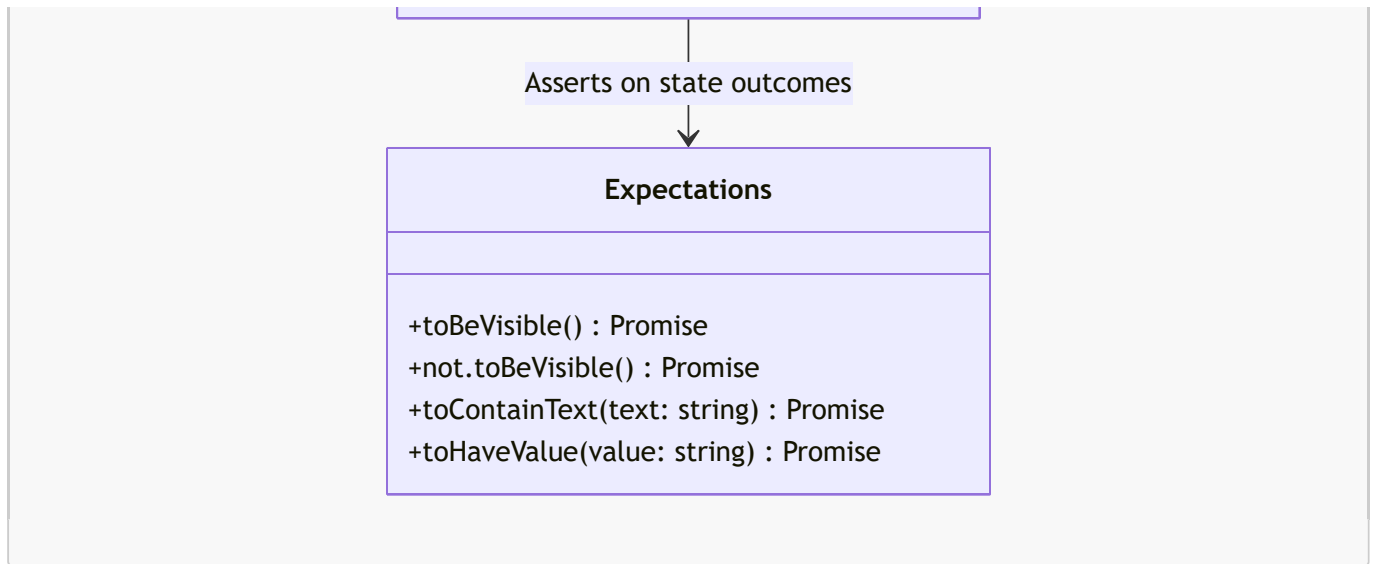
This document provides a highly detailed, precise set of UML diagrams and structural domain models mapping the actual Playwright E2E automation tests ([e2e/auth-and-movies.spec.ts](#)) generated for the Movie Review & Finder application.

1. Playwright Test Suite Structural Class Diagram

This class-level structural diagram represents the design, modules, and relationships within our actual E2E test file ([e2e/auth-and-movies.spec.ts](#)), including the helper functions, Page object representations, selector handles, and expectations.

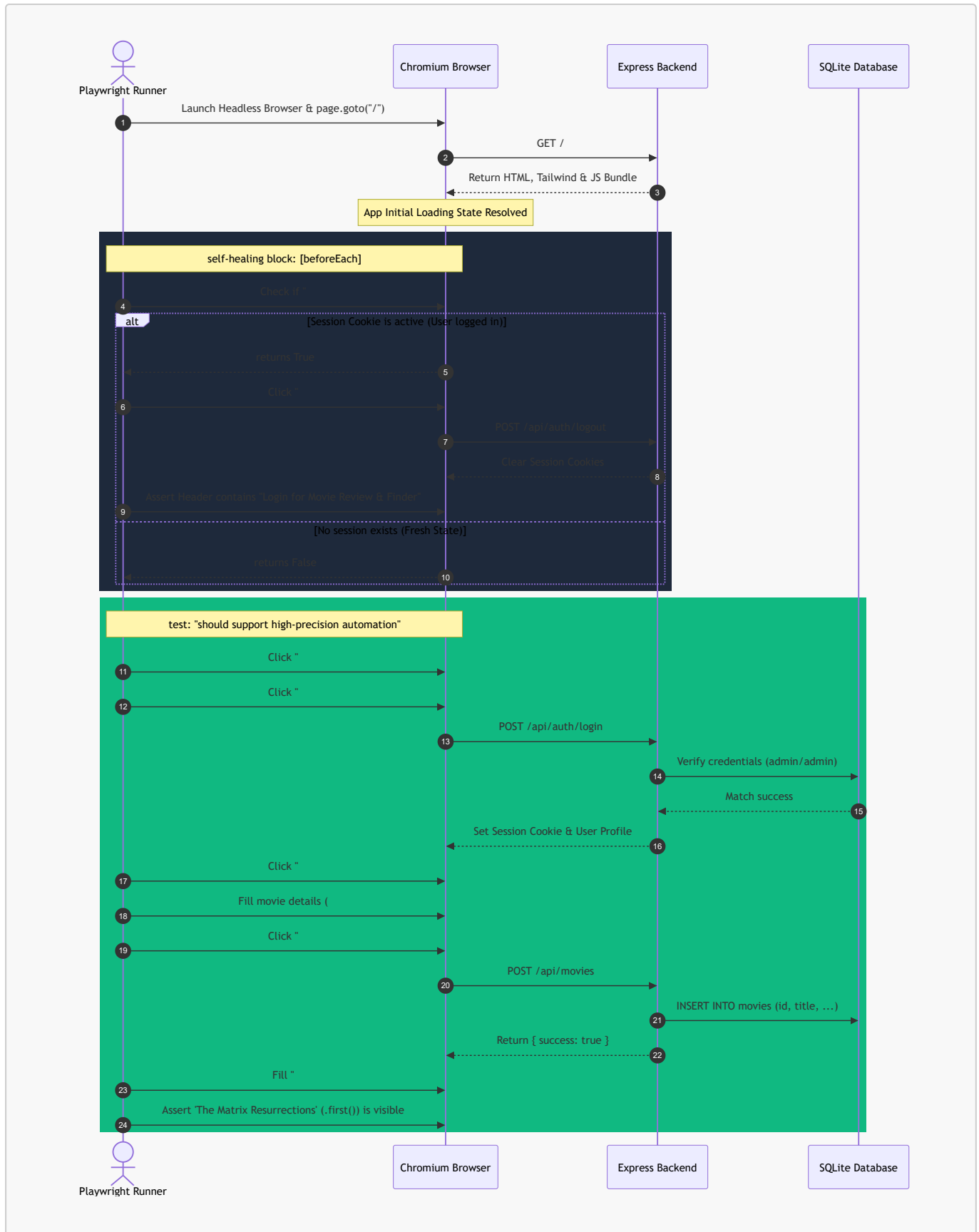






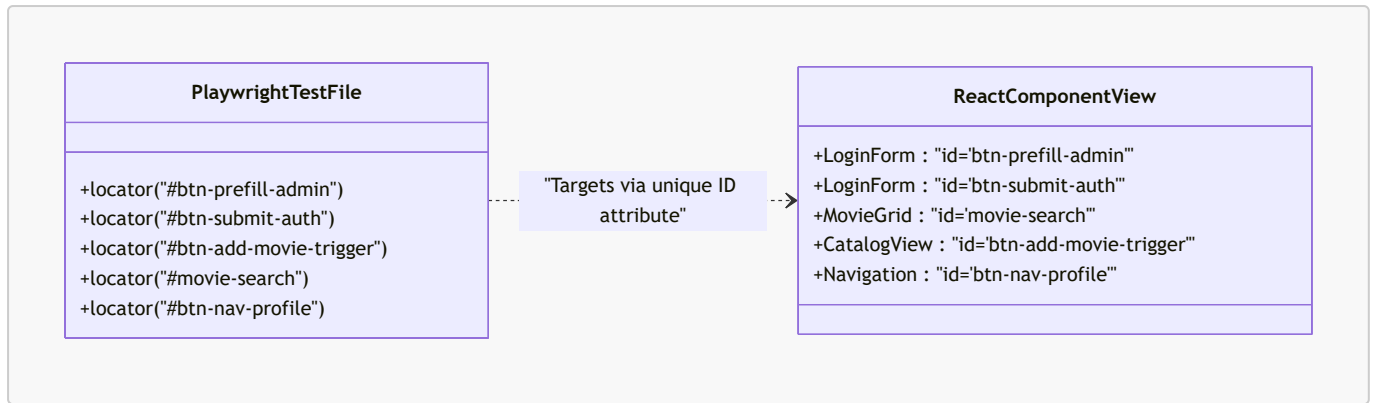
2. Test Execution & Self-Healing Sequence Diagram

This sequence diagram illustrates the step-by-step execution path of our tests, detailing how Playwright starts, handles active session cleanup (**beforeEach** self-healing logic), performs targeted interactions, and evaluates conditions.



3. High-Precision Selector Domain Model

This model demonstrates the direct, strict mapping between the automation test locators used in `auth-and-movies.spec.ts` and the exact DOM elements of our React front-end. It emphasizes how custom test attributes protect tests against selector breakage.



4. Key Automation Capabilities Mapped

Our automation tests are structured to cover these exact mechanics:

1. State Setup & Reset:

- `page.goto('/')` ensures routing starts at a clean origin.
- An active check on `#btn-logout` terminates any lingering sessions, achieving **test isolation**.

2. Element Queries & Scope Restriction:

- `page.locator('form button[type="submit"]')` scopes target buttons strictly to form elements.
- `page.getByRole('button', { name: 'Edit Profile' })` utilizes modern accessibility (ARIA) tree selectors.

3. Ambiguity Handling:

- `.first()` (e.g. `page.locator('text=The Matrix Resurrections').first()`) prevents strictness violations when multiple objects match queries.